Lab Exercise 13- Managing Namespaces in Kubernetes

Step 1: Understand Namespaces

Namespaces provide a mechanism for scoping resources in a cluster. Namespaces can be used to:

- Create environments for different applications or teams.
- Apply policies like resource quotas or network policies on a per-namespace basis.
- Separate operational environments (like development and production).

Step 2: List Existing Namespaces

To list all the namespaces in your Kubernetes cluster:

```
kubectl get namespaces
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl get namespaces

NAME                   STATUS   AGE
default                Active   15h
kube-node-lease        Active   15h
kube-public            Active   15h
kube-system            Active   15h
kubernetes-dashboard   Active   15h
mohdanas@Mohds-MacBook-Air Desktop %
```

You will typically see default namespaces like default, kube-system, and kube-public.

Step 3: Create a Namespace

You can create a namespace using a YAML file or directly with the kubectl command.

Using YAML File

Create a file named my-namespace.yaml with the following content:

```
apiVersion: v1
kind: Namespace
metadata:
  name: my-namespace
```

Apply this YAML to create the namespace:

```
kubectl apply -f my-namespace.yaml
```

```
[mohdanas@Mohds-MacBook-Air Desktop % vim my-namespace.yaml
[mohdanas@Mohds-MacBook-Air Desktop % kubectl apply -f my-namespace.yaml

 namespace/my-namespace created
 mohdanas@Mohds-MacBook-Air Desktop %
```

Using kubectl Command

Alternatively, create a namespace using the kubectl command:

```
kubectl create namespace my-namespace
```

Verify that the namespace is created:

```
kubectl get namespaces
```

You should see my-namespace listed in the output.

```
[mohdanas@Mohds-MacBook-Air Desktop % kubectl get namespaces

 NAME                   STATUS   AGE
 default                Active   15h
 kube-node-lease        Active   15h
 kube-public            Active   15h
 kube-system            Active   15h
 kubernetes-dashboard   Active   15h
 my-namespace           Active   2m10s
 mohdanas@Mohds-MacBook-Air Desktop %
```

Step 4: Deploy Resources in a Namespace

Create resources such as Pods, Services, or Deployments within the new namespace.

Deploy a Pod in the Namespace

Create a YAML file named nginx-pod.yaml with the following content:

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  namespace: my-namespace
spec:
  containers:
  - name: nginx
    image: nginx:latest
    ports:
```

```
  - containerPort: 80
```

Apply this YAML to create the Pod:

```
kubectl apply -f nginx-pod.yaml
```

Check the status of the Pod within the namespace:

```
kubectl get pods -n my-namespace
```

To describe the Pod and see detailed information:

```
kubectl describe pod nginx-pod -n my-namespace
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl apply -f nginx-pod.yaml
pod/nginx-pod created
mohdanas@Mohds-MacBook-Air Desktop % kubectl get pods -n my-namespace

NAME        READY   STATUS              RESTARTS   AGE
nginx-pod   0/1     ContainerCreating   0          6s
mohdanas@Mohds-MacBook-Air Desktop %
```

Create a Service in the Namespace

Create a YAML file named **nginx-service.yaml** with the following content:

```
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
```

```
  namespace: my-namespace
spec:
  selector:
    app: nginx-pod
  ports:
  - protocol: TCP
    port: 80
    targetPort: 80
  type: ClusterIP
```

Apply this YAML to create the Service:

```
kubectl apply -f nginx-service.yaml
```

```
mohdanas@Mohds-MacBook-Air Desktop % vim nginx-service.yaml
mohdanas@Mohds-MacBook-Air Desktop % kubectl apply -f nginx-service.yaml

service/nginx-service created
mohdanas@Mohds-MacBook-Air Desktop %
```

Check the status of the Service within the namespace:

```
kubectl get services -n my-namespace
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl get services -n my-namespace
NAME            TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)   AGE
nginx-service   ClusterIP   10.109.58.50   <none>        80/TCP    13s
mohdanas@Mohds-MacBook-Air Desktop %
```

To describe the Service and see detailed information:

```
kubectl describe service nginx-service -n my-namespace
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl describe service nginx-service -n my-namespace

Name:                     nginx-service
Namespace:                my-namespace
Labels:                   <none>
Annotations:              <none>
Selector:                 app=nginx-pod
Type:                     ClusterIP
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       10.109.58.50
IPs:                      10.109.58.50
Port:                     <unset>  80/TCP
TargetPort:               80/TCP
Endpoints:
Session Affinity:         None
Internal Traffic Policy:  Cluster
Events:                   <none>
```

Step 5: Switching Context Between Namespaces

When working with multiple namespaces, you can specify the namespace in kubectl commands or switch the default context.

Specify Namespace in Commands

You can specify the namespace directly in kubectl commands using the -n or --namespace flag:

```
kubectl get pods -n my-namespace
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl get pods -n my-namespace

NAME        READY   STATUS    RESTARTS   AGE
nginx-pod   1/1     Running   0          3m6s
```

Set Default Namespace for kubectl Commands

To avoid specifying the namespace every time, you can set the default namespace for the current context:

```
kubectl config set-context --current --namespace=my-namespace
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl config set-context --current --namespace=my-namespace
Context "minikube" modified.
```

Verify the current context's namespace:

```
kubectl config view --minify | grep namespace
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl config view --minify | grep namespace

    namespace: my-namespace
```

Step 6: Clean Up Resources

To delete the resources and the namespace you created:

```
kubectl delete -f nginx-pod.yaml
kubectl delete -f nginx-service.yaml
kubectl delete namespace my-namespace
```

Ensure that the namespace and all its resources are deleted:

```
kubectl get namespaces
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl delete -f nginx-pod.yaml

pod "nginx-pod" deleted
mohdanas@Mohds-MacBook-Air Desktop % kubectl delete -f nginx-service.yaml

service "nginx-service" deleted
mohdanas@Mohds-MacBook-Air Desktop % kubectl delete namespace my-namespace

namespace "my-namespace" deleted
mohdanas@Mohds-MacBook-Air Desktop % kubectl get namespaces
NAME                   STATUS   AGE
default                Active   15h
kube-node-lease        Active   15h
kube-public            Active   15h
kube-system            Active   15h
kubernetes-dashboard   Active   15h
mohdanas@Mohds-MacBook-Air Desktop %
```